

Unified HDR reconstruction from raw CFA data

Joel Kronander[†], Stefan Gustavson[†], Gerhard Bonnet[‡], Jonas Unger[†]

[†] Linköping University [‡] SpheronVR AG

joel.kronander@liu.se

Abstract

HDR reconstruction from multiple exposures poses several challenges. Previous HDR reconstruction techniques have considered debayering, denoising, resampling (alignment) and exposure fusion in several steps. We instead present a unifying approach, performing HDR assembly directly from raw sensor data in a single processing operation. Our algorithm includes a spatially adaptive HDR reconstruction based on fitting local polynomial approximations to observed sensor data, using a localized likelihood approach incorporating spatially varying sensor noise. We also present a realistic camera noise model adapted to HDR video. The method allows reconstruction to an arbitrary resolution and output mapping. We present an implementation in CUDA and show real-time performance for an experimental 4 Mpixel multi-sensor HDR video system. We further show that our algorithm has clear advantages over state-of-the-art methods, both in terms of flexibility and reconstruction quality.

1. Introduction

HDR imaging enables a wide range of new applications in image processing, computer graphics and cinematography. It is a field currently in rapid development, spurred by dramatic improvements over the last few years in digital image sensors and camera systems. However, state-of-the-art sensor technology is still unable to capture the full dynamic range in a natural scene with enough precision in a single exposure. There is thus a need to develop new HDR reconstruction algorithms that are general, flexible, of high quality and fast enough for routine use. In particular HDR video [17, 1, 35, 34, 32] involves large amounts of data, and real-time processing is required for a proper viewfinder and other interactive feedback to the camera operator.

Robust HDR image reconstruction operating on the raw output from the sensor(s) should: perform conversion from color filter array (CFA) sampled data to color images (*demosaicing*), allow and compensate for misalignments between the exposures (*resampling*) and fuse LDR images

from any set of varying exposures into a seamless HDR image (*HDR assembly*).

The vast majority of recent HDR reconstruction methods, e.g. [2, 10, 32], have considered resampling and demosaicing as separate steps to be performed either before or after HDR assembly. There are several problems with this approach. Demosaicing before HDR assembly, e.g. performing classic exposure bracketing with a regular LDR camera as in [6], causes problems with bad or missing data around saturated pixels. This is especially problematic as color channels saturate at different levels. On the other hand, demosaicing after HDR assembly, as performed by [2, 32] causes problems with blur and ghosting unless the sensors are perfectly aligned. For multi-sensor systems using high resolution sensors, it is problematic and costly, sometimes even impossible, to exactly match the CFA patterns of the sensors. Tocci *et al.* [32] report misalignment errors around the size of a pixel despite considerable alignment efforts. For exposure bracketing sequences, misalignment may result from camera shake. Hence, overlapping pixels from different sensors might not be of the same color. Furthermore, treating demosaicing, resampling and HDR assembly as separate problems makes it difficult to analyze and incorporate sensor noise and other pixel characteristics in the HDR reconstruction in a coherent and formal manner. Recently some methods performing joint denoising and demosaicing, joint super-resolution and HDR assembly and joint demosaicing and HDR assembly have been proposed, see section 2.

The main contribution of this article is a new approach addressing all of these problems. The basis of the work is an adaptive spatial and cross-sensor filtering using a local polynomial approximation (LPA) based on a noise-aware maximum likelihood estimation considering the heteroskedastic sensor noise. During reconstruction, we consider the entire set of valid, non-saturated LDR samples available around a certain point, and perform CFA interpolation, resampling and HDR assembly in a single operation. This leads to a reconstruction framework that is flexible, robust, and applicable to a wide range of multi-sensor and multi-exposure capture systems. A key feature of our algorithm is its com-

putational speed, making it applicable also to real-time assembly of high resolution HDR video. Our CUDA implementation on a Geforce 680 GPU is capable of reconstructing 4 Mpixel HDR frames at a sustained rate of 25 fps. As a part of the framework we have also developed a sensor noise model tailored to our example applications. To reconstruct color images, we reconstruct each color channel independently. Thus, unlike more advanced demosaicing methods [11], our current implementation does not utilize cross-correlation between channels to improve the estimate.

We demonstrate the benefits of our approach on simulated data from a range of sensor setups, including the recently proposed hardware setup of Tocci *et al.* [32], as well as on data obtained from our own experimental multi-sensor HDR video camera. The supplementary material attached to this paper contains all HDR images used in the comparisons as well as processed video examples.

2. Related Work

The algorithms presented in this paper is related to a large body of previous work, ranging from HDR capture and reconstruction, e.g. [29, 27], to theory and algorithms for accurate image reconstruction and image fusion, e.g. [5]. In this section, we give an overview of the previous work most closely related to the methods proposed in this paper.

Non-parametric Image Processing - The last decade has seen an increased popularity of image processing operations using locally adaptive filter weights, for applications in e.g. interpolation, denoising and upsampling. Examples include normalized convolution [20], the bilateral filter [33], and moving least squares [21]. Recently, deep connections have been shown [30, 25] between these methods and traditional non-parametric statistics [22]. In this paper, we fit Local Polynomial Approximations (LPA) [5] to irregularly distributed samples around output pixels using a localized maximum likelihood estimation [31] to incorporate the heterogeneous noise of the samples.

HDR fusion - The most common method for HDR reconstruction from a set of differently exposed Low Dynamic Range (LDR) images is to compute a per-pixel weighted average of the LDR measurements. The weights, often based on heuristics, are chosen to suppress image noise and to remove saturated values from processing [23, 6, 3, 18]. Weight functions can also be based on more sophisticated camera noise models. Mitsunaga and Nayar [26] derived a weight function that maximizes SNR assuming signal-independent additive noise. Kirk and Andersen [19] derived a weight function inversely proportional to the temporal variance of the digital LDR values. Granados *et al.* [10] later extended this approach to include both spatial and temporal camera noise. While most previous methods consider only a single pixel at a time from each LDR exposure, Tocci *et al.* [32] presented an algorithm that

incorporates a neighbourhood of LDR samples in the reconstruction. This method helps smooth the transition regions between sensors and performs well for sensors with large exposure differences, over 3 *f*-stops apart.

Image Fusion - When multiple images of the same scene are captured within sub-pixel displacements of each other, super-resolution reconstruction methods can be applied. A class of computationally simple super-resolution techniques are those based on interpolation in a common high resolution grid [24]. These methods typically first use some registration to put observed images in a common high resolution reference frame. The high-resolution image is then estimated in a least squares sense given a local neighbourhood of observed pixels [8, 13]. As noted in previous work, super-resolution and HDR imaging complement each other. Super-resolution increases the resolution in the spatial domain, and HDR increases the radiometric range and resolution. Several methods have also been proposed to incorporate both enhancements in the same processing framework [12, 37]. Most modern camera systems acquire color images using a sensor covered by a color filter array, and estimates the final color vector for each pixel by demosaicing, interpolation of neighboring color samples. For an overview of demosaicing, see Gunturk *et al.* [11]. Joint denoising and demosaicing have been considered in previous work [16, 15]. Farsiu *et al.* [7] proposed a framework for joint super-resolution and demosaicing based on a maximum a posteriori (MAP) optimization over all pixels in the output image, considering priors that enforce correlation between color channels. A similar approach was taken in [4] for HDR imaging with joint demosaicing for sensors with non-destructive readout (ie perfect alignment between consecutive readouts/frames). While a global MAP optimization could possibly also be used for joint realignment, HDR fusion and demosaicing, a global optimization is computationally demanding and does not straightforwardly allow for parallel processing.

To the authors knowledge, no previous approach has considered HDR assembly with joint realignment, noise reduction and CFA interpolation. While our method is not designed to provide optimal results with respect to denoising, demosaicing or super-resolution alone, we take a unified approach based on a model of sensor noise, providing a flexible framework which is straightforward to implement and use. Our method is also well suited to the needs of a versatile high dynamic range video system, by allowing an arbitrary resolution for the reconstruction and fast parallel processing.

3. Image Formation Model

Given a set of N differently exposed CFA images, I_s $s = 1 \dots N$, the goal is to reconstruct an HDR frame, F . The images, I_s , could for example be separate sensor im-

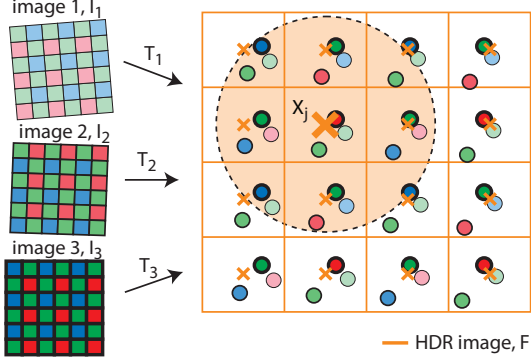


Figure 1: HDR reconstruction is performed in a virtual reference space with arbitrary resolution and mapping (orange grid). Measured image pixels are mapped to this grid by transformations T_s . A pixel $F(X_j)$ in the HDR image (orange cross) is estimated by a local polynomial approximation of nearby samples (within the shaded circle). Saturated and defective samples are discarded, and samples near the black level, local outliers and samples farther from the reconstruction point contribute less to the radiance estimate.

ages from a digital multi-sensor imaging system, or a set of still images captured using multi-exposure techniques. We make the following assumptions about the imaging system:

- For multi-sensor systems, the sensors are synchronized and use the same exposure time, or the scene is assumed to be static. For multi-exposure systems, the scene is assumed to be static, and the camera is assumed to be static enough to avoid parallax.
- The sensor images view the scene through a common optical system implying same focus, point spread function, and vignetting for all sensor images.
- Geometric misalignments between the sensor images can be described by some 2-D transformation T_s , so that they can be registered to a common frame of reference.
- The parameters that are allowed to vary between the sensor images are: the exposure time t_s , the gain setting g_s , and an optional exposure scaling coefficient, n_s .

4. Reconstruction Algorithm

Each sensor image, I_s , samples the incident radiant power, f , at a set of discrete pixel locations. Using a linear index i for pixels in each sensor image, we define the measured digital sample value at a pixel i in image I_s as $y_{s,i}$. The samples, $y_{s,i}$, contain measurement noise that is dependent on the input signal, f , as well as the sensor characteristics. After performing radiometric and geometric calibration of the sensors, the final HDR image, F , is estimated in one joint filtering operation directly from raw data.

4.1. Radiometric Calibration

The (noisy) digital pixel values $y_{s,i}$ in each input image are converted to radiant power estimates $\hat{f}_{s,i}$ by subtracting a bias frame and accounting for the exposure settings (exposure time, exposure scaling, global gain) and per-pixel gain variations. Using a radiometric camera noise model, the variance of the radiant power estimates is also estimated, $\sigma_{\hat{f}_{s,i}}^2$. Similarly to previous work [10] we assume the noise to follow a Gaussian distribution. More details on this assumption are given in Section 5, where we describe our radiometric noise model adapted for HDR video. Our framework can use any radiometric noise model assuming an approximately Gaussian noise distribution.

4.2. Geometric Alignment

The HDR reconstruction is performed in a virtual reference space, corresponding to a virtual sensor placed somewhere in the focal plane. The virtual sensor dimensions are chosen to reflect the desired output frame, unconstrained by the resolution of the input frames. Using a geometric calibration procedure, an affine transform, T_s , is established for each sensor which maps sensor pixel coordinates, $x_{s,i}$, to the coordinates of the reference output coordinate system, $X_{s,i} = T_s(x_{s,i})$. In general, the transformed input pixel coordinates, $X_{s,i}$, are not integer-valued in the output image coordinates, and for general affine transformations the sample locations $X_{s,i}$ will become irregularly positioned. An example for three sensor images is shown in Figure 1.

4.3. LPA Reconstruction

To reconstruct the HDR output frame, F , we estimate the radiant power for each pixel in the virtual sensor separately in a non-parametric fashion using the transformed samples $\hat{f}_{s,i}(X_{s,i})$. For each output pixel j , with integer valued, regularly spaced coordinates X_j , we fit a polynomial model to observed samples $\hat{f}_{s,i}(X_{s,i})$ in a local neighbourhood. We perform the reconstruction of a single color channel, $c = \{R|G|B\}$, independently of the other color channels. Thus, unlike more advanced demosaicing methods [11], our current implementation does not utilize cross-correlation between channels to improve the estimate.

A local polynomial model of order M predicts the signal value at a point X_i in the neighbourhood of the fitted output location, X_j as

$$z(X_i, X_j, C) = C^T \bar{\phi}(X_j - X_i) \quad (1)$$

$$\bar{\phi}(x) = [\phi_0(x), \phi_1(x), \dots, \phi_M(x)]^T \quad (2)$$

$$C = (C_0, C_1, \dots, C_M)^T \quad (3)$$

where $\bar{\phi}(X_j)$ is a vector of 2D polynomials and C are the polynomial coefficients. For example, for $M = 0$,

$[\phi_0(x) = 1]$, and for $M = 1$, $[\phi_0(x) = 1, \phi_1(x) = x_1, \phi_2(x) = x_2]$ where x_1 denotes the first and x_2 denotes the second coordinate of the point. This polynomial expansion is equivalent to locally fitting a truncated Taylor expansion in the vicinity of each output pixel.

To estimate the coefficients, C , of the local polynomial model, we maximize a localized likelihood function [31] defined using a smoothing window, $\mathcal{W}_h(X) = \frac{1}{h^2} \mathcal{W}(\frac{X}{h})$, centered around the output location, X_j . Where $\mathcal{W}$ is any normalized window function, e.g. a Gaussian, and h is a scale parameter that determines the size of the window. We discuss the choice of the window function $\mathcal{W}$ and the scale parameter h in section 4.3.1. To lessen the notational burden, the radiometrically calibrated pixel samples in the neighborhood, $\{\hat{f}_{s,i}(X_{s,i}) : X_{s,i} \in \text{supp}(\mathcal{W}_h(X))\}$, are denoted below by f_k with a linear index $k = 1 \dots K$. Assuming that the estimates f_k , are normally distributed, see Section 5, the local log likelihood of the polynomial expansion centered at X_j is given by

$$L(X_j, C) = \sum_k \log(N(f_k | z(X_k, X_j, C), \hat{\sigma}_{f_k}^2) w_k) \\ = -\frac{1}{2} \sum_k (f_k - C^T \bar{\phi}(X_j - X_k))^2 \frac{w_k}{\hat{\sigma}_{f_k}} + R$$

where R represents terms independent of C and w_k is the smoothing weight given by the window function, $\mathcal{W}_h(X_k)$, evaluated at sample X_k . The polynomial coefficients, C , maximizing the local likelihood function are found by the weighted least squares estimate

$$\tilde{C} = \underset{C \in \mathcal{R}^M}{\text{argmax}} (L(X_j, C)) \quad (4)$$

$$= (\Phi^T W \Phi)^{-1} \Phi^T W \bar{f} \quad (5)$$

where $\Phi = [\bar{\phi}(X_j - X_1)^T, \bar{\phi}(X_j - X_2)^T \dots \bar{\phi}(X_j - X_K)^T]^T$ is a $K \times M + 1$ design matrix, $W = \text{diag}[\frac{w_1}{\hat{\sigma}_{f_1}}, \frac{w_2}{\hat{\sigma}_{f_2}}, \dots, \frac{w_K}{\hat{\sigma}_{f_K}}]$ is a $K \times K$ diagonal weight matrix, and $\bar{f} = [f_1, f_2, \dots, f_K]^T$.

Using the fitted polynomial coefficients, $\tilde{C}$, we can estimate the radiant power at the virtual sensor pixel as

$$\hat{f}(X_j) = z(X_j, X_j, \tilde{C}) = \tilde{C}^T \phi(0) = \tilde{C}_0 \quad (6)$$

For polynomial models of order $M > 0$, the derivatives, $\hat{f}^{1,0}(X_j)$ and $\hat{f}^{0,1}(X_j)$ at the output location, X_j , can be estimated using the higher order coefficients $\tilde{C}_1$ and $\tilde{C}_2$ [5].

4.3.1 Parameters

The expected mean square error of the reconstructed image depends on a trade-off between bias and variance of the estimate. This trade-off is determined by: the order of the polynomial basis, M , the window function, $\mathcal{W}$, and the

scale parameter, h . Finding the optimal balance between these parameters is, in the general case, very difficult and still an open problem in non-parametric statistics [22]. Instead of relying on automatic selection of these parameters, we treat them as user parameters that are set depending on the sensor configuration and scene properties.

Polynomial Order - Using a piecewise constant polynomial, $M = 0$, the estimator corresponds to an ordinary locally weighted average of neighboring sensor observations

$$\hat{f}(X_j) = \frac{\sum_k \frac{w_k}{\hat{\sigma}_{f_k}} f_k}{\sum_k \frac{w_k}{\hat{\sigma}_{f_k}}} \quad (7)$$

However, locally weighted averages can exhibit severe bias at image boundaries and in regions around sensor saturation, due to the asymmetry of the number of available sensor measurements in these locations. By instead fitting a linear polynomial (a plane, $M = 1$), the bias can be reduced significantly. Introducing higher order polynomials is possible, but may lead to increased variance in the estimates. In this work, we only consider $M \leq 2$.

Window and scale selection - We show results using a Gaussian window function, $\mathcal{W}(X) = \frac{1}{\sqrt{2\pi}} e^{-\frac{X^2}{2}}$, with the Euclidean distance to the reconstruction point X_j as argument. Other symmetric smooth window functions could also be used, e.g. a Epanechnikov or a Tricube window [5]. To reconstruct sharp images in low noise conditions, we generally optimize for small window supports, choosing the smallest possible scale parameter, h , such that $(\Phi^T W \Phi)^{-1}$ is invertible (full rank) for all X_j . However, if the captured images exhibit severe noise, a larger scale parameter may be beneficial. For Bayer CFA sensors, we consider different scales for the green and red/blue color channels, $h_G = \frac{h_{R,B}}{\sqrt{2}}$, as there are more green samples per unit area.

4.4. CUDA Implementation

Our software implementation of the algorithm required several seconds for processing of reasonably sized frames. To attain real time processing speed for video data, we implemented the algorithm in CUDA. Two CUDA kernels are executed per frame. First, the raw 12-bit sensor images are converted to a half-float radiant power estimate, $\hat{f}_{s,i}$, and a variance estimate, $\hat{\sigma}_{f_{s,i}}$. Second, for each pixel in the output image, all samples overlapping the observation window are read from global GPU memory, their corresponding weights are computed, and the weighted least squares estimate of Equation (5) is calculated. To increase memory transfer efficiency, page-locked and write-combined memory is used. Data is transferred to the GPU for kernel execution and then back to the CPU for disk storage. Two CUDA streams are used to allow for simultaneous data transfer and kernel execution. If the sensors are aligned or related by purely translational transforms, and if the output resolution matches the

input resolution, the spatial arrangement of samples in each observation window will be invariant across the image, and the window weights w_i can be precomputed once. This results in a $2\text{-}3\times$ speedup for a local constant fit $M = 0$.

5. Sensor noise model

In this section we describe a radiometric camera noise model assuming a linear digital response. We model the noise using a radiometric camera model inspired by previous methods considering radiometric camera calibration [28, 9] and optimal HDR capture methods [10, 14].

Each observed digital pixel value, $y_{s,i}$, is obtained using an exposure time, t_s , and an exposure scaling, n_s that is assumed constant over the image I_s . The pixel response is a measurement of the radiant power reaching the image sensor, which we for convenience express as the number of photo-induced electrons collected per unit time, $f_{s,i}$. The accumulated number of photoelectrons, $e_{s,i}$, collected at the pixel during the exposure time, t_s , follows a Poisson distribution with expected value

$$E[e_{s,i}] = t_s(a_{s,i}n_s f_{s,i}) \quad (8)$$

and variance $Var[e_{s,i}] = E[e_{s,i}]$ where $a_{s,i}$ is a per pixel factor due to non-uniform photo-response (photosensitive area). The recorded digital value, $y_{s,i}$, is also affected by the sensor gain and signal independent readout noise,

$$y_{s,i} = g_s(e_{s,i}) + r_{s,i}(g, t) \quad (9)$$

where g_s is the amplifier/sensor gain (proportional to ISO) and $r_{s,i}(g, t)$ is the readout noise, that can be dependent on both the gain, g_s , and the exposure time, t_s , settings of the sensor. For the Kodak KAI-04050 sensor used in our experimental setup, see Section 7, we have found that the readout noise can be modeled by a pre-amplifier noise, $p_{s,i}$, and a post amplifier quantization noise, q_s [36]. We can thus form a parametric model of the readout noise as $d_{s,i}(g, t) = g(p_{s,i}) + q_s$. However, for other image sensors other parametric models could be more appropriate, e.g. Hasinoff *et al.* [14] models the quantization noise with a more general post-amplifier noise. As we focus on video applications with frame rates of 25 fps or more, the dark current noise is neglected

Before reconstruction of the HDR output frame, each digital pixel value, $y_{s,i}$, is independently transformed to an estimate of the number of photoelectrons reaching the pixel per unit time, $\hat{f}_{s,i}$. To compensate for the readout noise bias (blacklevel), $E[d_{s,i}(g, t)]$, we subtract a *bias frame*, $b_{s,i}$, from each observation. The bias frame is computed as the average of a large set of black images captured with the same camera settings as the observations but with the lens covered, so that no photons reach the sensor. The radiant

power $f_{s,i}$ can then be estimated as

$$\hat{f}_{s,i} = \frac{y_{s,i} - b_{s,i}}{g_s t_s a_{s,i} n_s} \quad (10)$$

5.1. Variance estimate

We assume $\hat{f}_{s,i}$ to follow a normal distribution with mean $f_{s,i}$ and standard deviation $\sigma_{\hat{f}}$. For low light levels, photon shot noise is generally dominated by signal independent readout noise approximately following a normal distribution, and for brighter areas, with a high photon count, the Poisson distributed shot noise is well approximated by a normal distribution. Assuming no saturated or clipped pixels values, the variance of $\hat{f}_{s,i}$ is given by

$$Var[\hat{f}_{s,i}] = \sigma_{\hat{f}}^2 = \frac{g_s^2(Var[e_{s,i}]) + Var[r_{s,i}(g, t)]}{g_s^2 t_s^2 a_{s,i}^2 n_s^2} \quad (11)$$

Due to the photon shot noise, to estimate, $Var[e_{s,i}]$, we should ideally know the true value of $f_{s,i}$, however in practice we are instead forced to use the (noisy) estimate, $\hat{f}_{s,i}$, obtained from equation (10). Thus we form an approximate variance estimate, $\hat{\sigma}_{\hat{f}_{s,i}}^2$, as

$$\hat{\sigma}_{\hat{f}_{s,i}}^2 = \frac{g_s^2 t_s a_{s,i} n_s \hat{f} + Var[r_{s,i}(g, t)]}{g_s^2 t_s^2 a_{s,i}^2 n_s^2} \quad (12)$$

The above analysis assumes non-saturated pixels. In practice, we exclude saturated pixels by comparing each digital pixel value, $y_{s,i}$, to a *saturation frame*. The model does not include the effect of pixel cross-talk, and the variances, $\sigma_{\hat{f}_{s,i}}^2$, are assumed to be independent of each other.

5.2. Parameter calibration

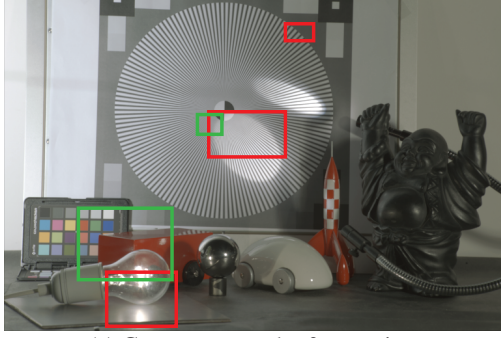
The exposure scaling, n_s , and per-pixel non uniformity, $a_{s,i}$, can in practice be estimated using a flat field image computed as the average over a large sequence of images. Choosing one sensor as reference, the exposure scaling, n_s , can be estimated as the spatial mean of the ratio, computed pixel wise between the radiant power estimates, Eq. 10, of the sensors. The non uniformity, $a_{s,i}$, can then be computed as the per pixel deviation from the exposure scaling, n_s .

The variance of the readout noise, $Var[r_{s,i}(g, t)]$ can be estimated, similarly to the bias frame, from a set of black images, captured with the same camera settings, (g, t) , as the observations but with the lens covered, so that no photons reach the sensor.

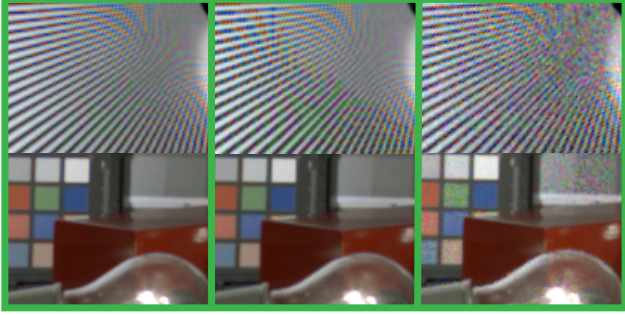
The sensor gain, g_s , can be calibrated using the relation,

$$\frac{Var[y_{s,i}] - Var[b_{s,i}]}{E[y_{s,i}] - E[b_{s,i}]} = \frac{g_s^2 Var[e_{s,i}]}{g_s E[e_{s,i}]} = g_s \quad (13)$$

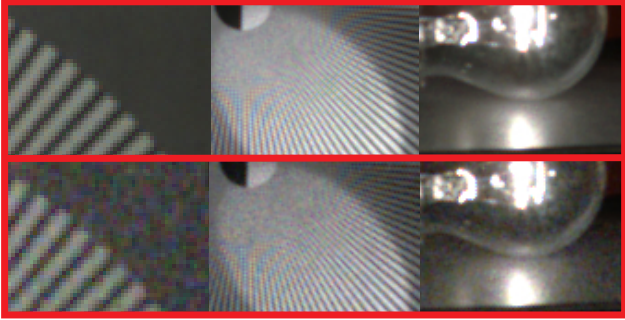
where the second equality follows from $e_{s,i}$ being Poisson distributed shot noise with $E[e_{s,i}] = Var[e_{s,i}]$. $E[y_{s,i}]$ and $E[b_{s,i}]$ can be estimated by averaged flat fields and the bias frame respectively, and $Var[b_{s,i}]$ as described above.



(a) Gamma mapped reference image



(b) Left to right: LPA (M=1), Debayer first, Tocci et al.



(c) Top: LPA (M=1), and bottom: Tocci et al.

Figure 2: (a) Tone-mapped HDR reference image. (b): Comparisons of reconstructions using LPA, a Debayer-first method and the method of Tocci *et al.* LPA reconstruction shows less color fringes and less overall noise. (c): Contrary to Tocci *et al.*, LPA has good inherent denoising abilities.

6. Algorithm evaluation

To evaluate our algorithm, we compare its performance against the recent real-time multi-sensor algorithm presented by Tocci *et al.* [32], which performs debayering after HDR reconstruction, and an algorithm performing debayering before HDR reconstruction and using linear radiometric weights (as proposed by Mitsunaga and Nayar [26] for a linear camera response curve). The latter represents the more traditional exposure bracketing approach, where demosaicing is performed before HDR assembly. This will be referred to as *Debayer-first*. As stated in the introduction, Debayer-first methods have a serious drawback in that many pixel measurements may be affected by nearby saturated or

dead pixels. For pixel-perfect sensor alignment, we also compare our method to the iterative algorithm proposed by Granados *et al.* [10].

We base several of our comparisons on image data generated from a camera simulation framework. The camera simulation framework allows us to generate images of arbitrary bit depth, sensor misalignments, exposure settings and noise characteristics according to the model described in Section 5. In the simulator, the scene is represented by a virtually noise free high resolution HDR image, meticulously generated from a set of exposures captured 1 *f*-stop apart covering the dynamic range of the real scene. Each exposure is computed as an average of 100 frames, in this case captured with a Canon 5D SLR camera. For our experiments, we simulate different multi-sensor setups with different sensor misalignments, T_s , and different noise characteristics, given the simulation parameters $g_s, n_s, t_s, Var[r_{s,i}]$.

Sensor misalignments - A general multi-sensor reconstruction algorithm requires robust handling of sub-pixel misalignments. Note that both the LPA reconstruction and the Debayer-first method can handle general misalignment errors, including rotations. However, methods performing demosaicing after resampling (alignment), e.g. Tocci *et al.* [32], cannot handle this correctly. In Figure 2b, LPA is compared to the method of Tocci *et al.* [32] and the Debayer-first method for sensors with translational misalignments. The three virtual sensors simulate the Kodak KAI-04050 sensor used in our experimental camera, see Section 7, but with exposure scaling coefficients $n_0 = 1$, $n_1 = 2^{3.69}$, $n_2 = 2^{7.78}$, simulating the setup presented by Tocci *et al.* The lowest exposed and the highest exposed sensors are perfectly aligned, while the middle sensor has a translational offset of 0.65, 0.9 in pixel units. This is similar to the alignment errors reported by Tocci *et al.* for their camera system.

The method of Tocci *et al.* copes rather well with translational misalignments, because each reconstructed pixel uses information from only one sensor. However, the reconstruction suffers from noise. Debayer-first suffers from color artifacts, seen as color fringes.

Perfect alignment - To evaluate the reconstruction quality of our method compared to Tocci *et al.* [32] we simulated four 10-bit sensors 3 *f*-stops apart, $n_0 = 1$, $n_1 = 2^3$, $n_2 = 2^6$, $n_3 = 2^9$, with a resolution of 512×764 pixels. The gain and readout noise were set to $g_s = 0.92$ and $d_{s,i} \sim \mathcal{N}(32, 18)$, corresponding to the measured sensor noise of a Canon PowerShot S5 [10]. The virtual sensors were perfectly aligned, with overlapping Bayer-CFA locations. In Figure 2c, the reconstruction of our method (LPA) is compared to the method presented in [32]. The LPA reconstruction exhibits low noise due to the adaptive filtering and data fusion between sensors, and reduces blur and color

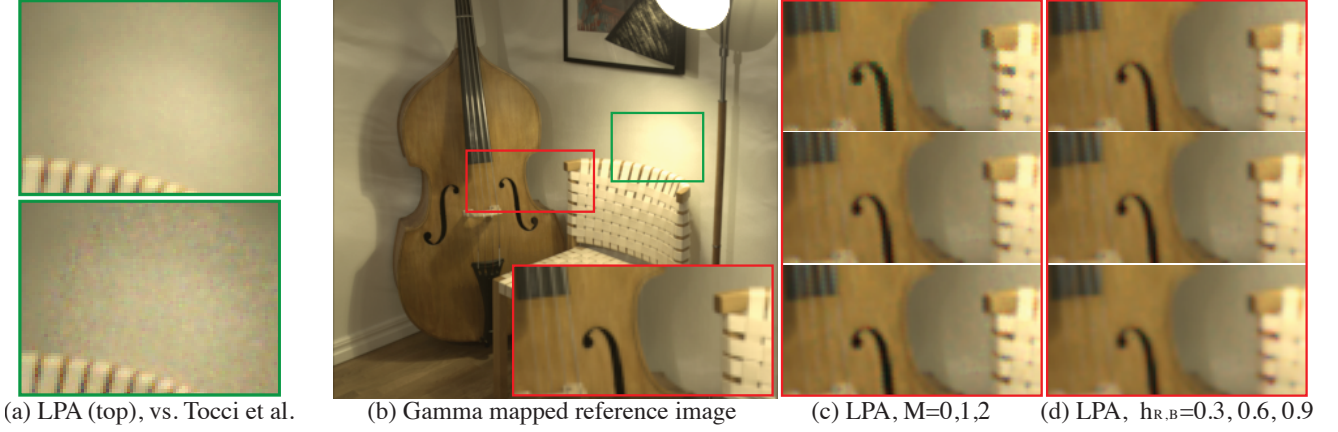


Figure 4: (a) Comparison between LPA reconstruction, $M = 1$, (top) and the method by Tocci et al. (bottom) from three simulated Canon 5D sensors 4 f -stops apart. (b) Tone-mapped HDR reference image. (c) LPA reconstructions of different polynomial order $M = 0, 1, 2$ with a fixed scale parameter, $h_G = 0.2$, $h_{R,B} = 0.4$. Using a constant polynomial ($M = 0$) yields blocky artifacts and color fringes close to sensor saturation and sharp edges. (d) LPA reconstructions, $M = 1$, for different scale parameters, $h_{R,B} = 0.3, 0.6, 0.9$. Using a low scale parameter leads to sharp reconstructions, but a larger scale parameter reduces image noise.

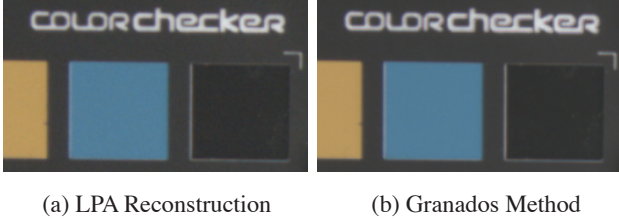


Figure 3: Reconstructed results from four perfectly aligned exposures 4 f -stops apart captured with a Canon 40D camera. (a) LPA Reconstruction and (b) the iterative method of Granados *et al.* [10].

artifacts by performing resampling and CFA interpolation jointly with reconstruction. This is expected, as the LPA algorithm uses a weighted least squares estimate based on all available sensor observations, while the method of Tocci *et al.* only considers observations from the highest exposed non-saturated sensor. LPA reconstruction also gracefully handles transitional regions around sensor saturation points, as a smoothing spatial filtering and a cross-sensor blending are both inherent in the reconstruction. This has the benefit of implicitly performing denoising.

Similar results can also be seen in Figure 4a, where the methods are compared using three virtual exposures, 4 f -stops apart, each simulating a Canon EOS 5D sensor set to ISO400, corresponding to a gain of $g_s = 0.23$ and a readout noise with variance 6.5 as reported by [10].

To compare our method to the algorithm proposed by Granados *et al.* [10] we captured four exposures four f -stops apart with a Canon 40D camera, with calibrated gain 1.24 (ISO 400), and readout noise variance 64.2 (14 bit sensor). For a fair comparison to our method we perform de-

mosacing (bilinear CFA interpolation) after HDR-assembly for Granados method, in contrast to the publicly available code that performs a crude nearest neighbour demosacing before HDR assembly. In Figure 3 it is visible that that Granados method performs slightly better, although the difference is not significant. Note that in contrast to the method of Granados *et al.* our method handles arbitrary misalignments between the sensors.

LPA parameters - Varying the polynomial order and the scale parameter of the local polynomial model imposes a trade-off between image sharpness, noise reduction and processing speed. Figure 4b and 4c show reconstructions from three aligned virtual Kodak KAI-04050 sensors 4 f -stops apart and a resolution of 512×341 pixels. The close-ups show how the image fidelity varies for different choices of polynomial order the M and the scale parameter h described in Section 4.3.1. Using a constant fit, $M = 0$, leads to unwanted block artifacts and color fringes, especially at edge features and in regions where one sensor is close to saturation. The difference between order $M = 1$ and $M = 2$ is more subtle, with only small improvements in visual quality. A low value for the scale parameter, h , increases image sharpness, while a large h reduces image noise.

7. Experimental validation

To evaluate the real world performance of our reconstruction algorithm, we employ a custom built multi-sensor system. Four high quality CCD sensors, Kodak KAI-04050 with a resolution of 2336×1752 pixels and RGB Bayer pattern CFA sampling, receive different amounts of light from an optical system with a common lens, a four-way beam splitter and four different ND filters, see Figure 5, top. The

relay lens, which extends the optical path length to make room for the beam splitters, was custom designed for this setup, but all other optics are standard, off-the-shelf components. Each sensor has 12 bits linear A/D conversion, and the exposure scalings of the four sensors, inferred by the ND-filters, cover a range of $1 : 2^{12}$. This yields a dynamic range equivalent to $12 + 12 = 24$ bits of linear resolution, commonly referred to as “24 f-stops” of dynamic range. The dynamic range can be extended further by varying the exposure times between the sensors. Each sensor is connected to a host computer through a CameraLink interface. The system allows for capture, processing and off-line storage of up to 32 frames per second at 4 Mpixels resolution, amounting to around 1 GiB/s per second of raw data.

Sensor misalignments were measured by imaging a checkerboard calibration target and computing the cross-sensor correlation. The misalignments were in our case translations (of a few pixels) and pure 2D rotations (in the order of fractions of a degree) and could be estimated with an accuracy within 0.1 pixels. The system was radiometrically calibrated as described in Section 5.2. The gain parameter was estimated for each pixel separately, and then spatially averaged to find the sensor gain $g_s = 0.27DV/e$. Readout noise was estimated for each pixel, with a cross-sensor spatial mean of $\hat{\mu}_{s,i} = 72$ e, and a standard deviation of $\hat{\sigma}_{s,i} \approx 11.8$ e.

Figure 5 shows an example frame captured using our experimental HDR-video system. Using our CUDA implementation with $M = 0$, and an output resolution of 2336×1752 pixels, the four input frames are processed at 26 fps using pre-computed weights on an Nvidia GForce 680 GPU. As can be seen in the tone-mapped image, which is in full resolution to enable zooming in during on-screen viewing, the small rotational misalignments present in the hardware setup are not visible even though the weights are pre-computed. This is partly due to the smoothing inherent in the LPA method. Reconstructing the same frame without pre-computed weights achieves an interactive speed of 7 fps. The performance scales linearly with the number of pixels, so the corresponding figures for reconstructing to HD resolution frames of 1920×1080 are 51 fps and 14 fps respectively.

8. Conclusions and Future Work

This paper presented a novel method for HDR reconstruction from multi-sensor and multi-exposure images that performs all steps in the traditional HDR imaging pipeline in a single step. The method is based on a sensor noise model adapted to multi-sensor systems and a novel HDR assembly algorithm based on local polynomial approximations. We also presented an overview of a novel HDR video camera and showed how our algorithm achieves real-time performance. As future work, we will further explore more

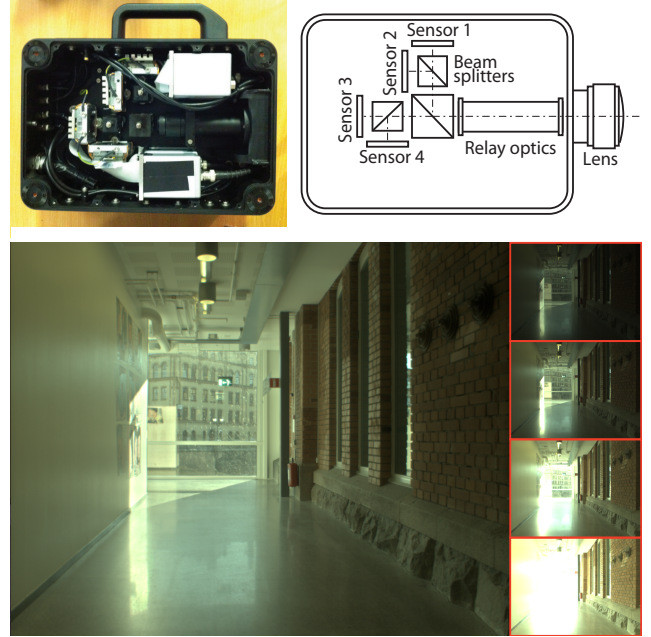


Figure 5: Top: An overview of our experimental HDR video camera. Bottom: a full resolution locally tone mapped frame from a video sequence captured with the HDR video camera. The image is reconstructed from four 4 Mpixel sensors at 26 fps. Bottom-right: Four gamma mapped images 2 f-stops apart from the same HDR-image.

advanced statistical denoising methods using anisotropic smoothing supports [5, 30].

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